

Math AI Lab



Inaugural meeting · Fall 2026

Dhruv Bhatia & Michael R. Zeng

ai.math.uw.edu

Spring Demo Day · June 8, 2026

PEOPLE

Math AI Lab



Michael R. Zeng

Co-Director



Dhruv Bhatia

Co-Director



**Giovanni
Inchiostro**

Faculty Advisor



Vasily Ilin

Founding Director



Jarod Alper

Founding Director

Faculty · graduate mentors · undergraduate researchers

Mathematics, AI, and learning together.

SPRING & SUMMER 2026

Recent work and events



UW Lean Hackathon

May 8–10 · Seattle

TAG-DS · August

Samarth Rao · spotlight



ICLR 2026 · ICML 2026

3 papers at ICLR

8 papers across ICML and its workshops

IEEE QCE · September

Andres Paz · StabilizerBench

Recent work

Semantic Search over 9 Million Mathematical Theorems

NeurIPS 2026, Evaluations and Datasets Track · Poster

Learning to Repair Lean Proofs from Compiler Feedback

ICLR 2026, VerifAI Workshop

CircuitBuilder: From Polynomials to Circuits via Reinforcement Learning

ICLR 2026, AI with Recursive Self-Improvement Workshop

Finite Étale Extensions of Local Rings Are Monogenic

Mathlib PR #37527 · Merged

mkOfAdjoinEqTop' for Adjoined Roots

Mathlib PR #36421 · Merged

DiScoFormer: Plug-In Density and Score Estimation with Transformers

ICML 2026 · Oral presentation

Formalizing Numerical Analysis: An Agent Pipeline and Quality Audit Beyond Kernel Acceptance

ICML 2026, AI for Math Workshop

Sorries Are Not the Hard Part: An Expert-Review Case Study of a Semi-Autonomous Formalization

ICML 2026, AI for Math Workshop

Semi-Autonomous Formalization of the Vlasov-Maxwell-Landau Equilibrium

ICML 2026, AI for Math Workshop

Learned Interventions Inside Lean 4's grind

ICML 2026, AI for Math Workshop

Does My Embedding Reflect That $A = B$? Evaluating Mathematical Equivalence in Embedding Models

TAG-DS 2026; ICML 2026, AI for Math Workshop · TAG-DS Spotlight

Multiplication Beyond Groups: Stratified Fourier Mechanisms in Transformer Circuits

ICML 2026, Mechanistic Interpretability Workshop · Spotlight

FactorLibrary: From Polynomials to Circuits via Recursive Subgoals

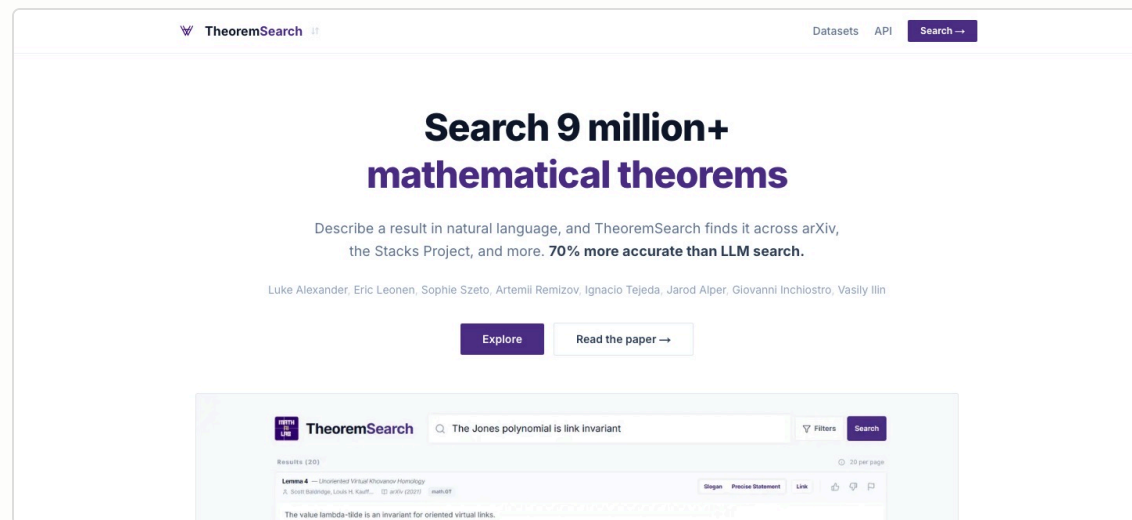
ICML 2026, AI for Math Workshop

StabilizerBench: A Benchmark for AI-Assisted Quantum Error Correction Circuit Synthesis

IEEE QCE 2026, QSYS track

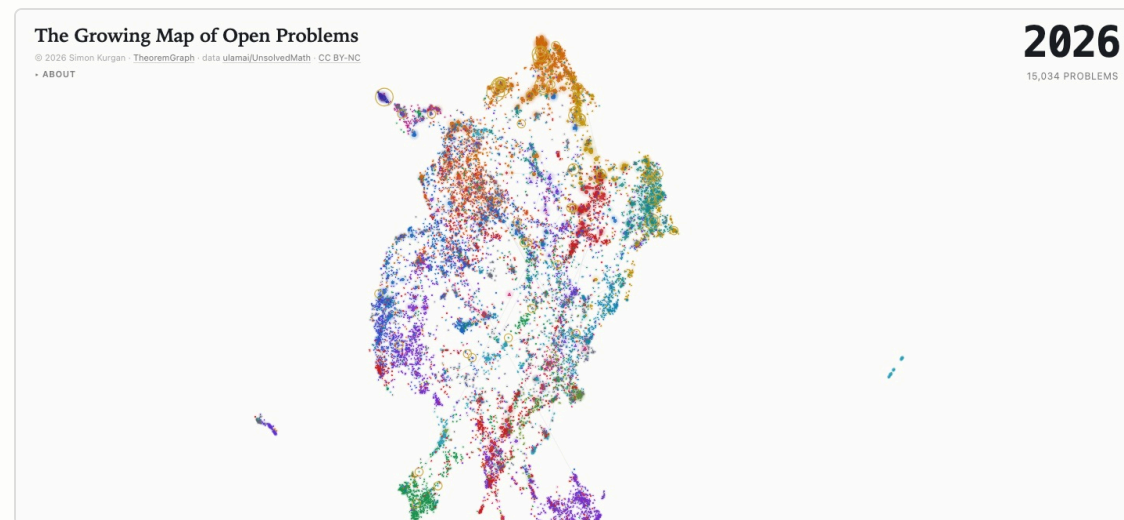
A FEW EXAMPLES

Recent work



TheoremSearch

Try TheoremSearch ↗



TheoremGraph

Explore the open-problems map ↗

Vlasov–Maxwell–Landau

Formalization in Lean

DiScoFormer

Density & score estimation

FactorLibrary

Arithmetic circuits

Fall 2026 projects

Autoresearch

6 projects

Mathematical machine learning & Interpretability

2 projects

Formalization & autoformalization

4 projects

Math education

2 projects

Autoresearch

AI-Assisted Proof Discovery in Inverse Boundary Problems

LEAD

Ruirui Wu

STUDENTS

Annie Feng

Kevin Wei

Kushal Kothapalli

Shiqi Yang

Harrison Bruechert

Mathematical Taste: Recognizing Progress Beyond Generation

LEAD

Zihong Lin

CO-MENTOR

Will Dudarov

STUDENTS

Ganesh Ravichandran

David An

Annie Cao

Duc Huy Nguyen

Bhuvan Gundela

Armen Agalyan

Autoresearch

Lean Refactor

LEAD

Evan Wang

STUDENTS

Kedar Chintalapati

David Javnozou

Chuhan Li

Evan Gubarev

Albilich v2: Math Metaharness

LEADS

Ting Gong · Michael R. Zeng

Dean Light · Michael Theologitis ·

Evan Wang

Metaharness meets Mondays, 3–4 p.m.

Autoresearch

Learning Conditional Return Laws from Financial Trading Signals

LEADS

Freda Zhang · Yue Wu

STUDENTS

Daran Xu

Belinda Krista

Rithikesh Muddana

Archit Ganapule

Chaoxiang Zhang

Formalizing the Scientific Method

LEADS

Eric Klavins · Bryan Boehnke

STUDENTS

Kavi Mahajan

Rohith Kala

Aiden Hoang

Formalization & autoformalization

Formalizing Algebra of Resultants and Determinantal Polynomials

LEADS

Joe Rogge · Shaoda Ji

STUDENTS

Grant Yang

Anders Lewis

Liron Shani

Nhi Bach

David Javnozon

Formalizing Modern Techniques in Probability Theory

LEAD

Brayden Letwin

CO-MENTOR

Will Dudarov

STUDENTS

Cosun Zhou

Aarush Sharma

Elisa Regueros

Owen Parks

Formalization & autoformalization

Autoformalizing Mathematical Benchmarks

LEAD

Theodore Meek

CONTINUING MEMBER

Michael R. Zeng, Pei Li, Simon Kurgan

Code Certifications

LEADS

Eric Klavins · Dhruv Bhatia

STUDENTS

Owen Fei

Sriram Marasanapalle-Kalle

Connor Zacharias

Yash Meher

Mathematical machine learning

PermuFormer 2

LEADS

Henry Kvinge

GRADUATE MENTORS

Michael R. Zeng · Junaid Hasan ·
Treanungkur Mal · Ron Cherny

STUDENTS

James Kwong	Yash Pratik Solanki
Emily Meng	Mayee Sun
Bhargava Perumalla	William Tran
Onkit Samanta	Xuanyu Yang
Allen (Ailun) Shen	Billy (Yinqi) Zhao

First meeting: Wed, Oct 7 · 4 p.m.

Mathematician's Copilot: Math2Vec

LEAD

Henry Kvinge

STUDENTS

Cecilia (Jiaying) Ye
Samarth Rao
Vaibhav Jasti
George Peykanu
Arnav Jain
Prithvi Aravind

First meeting: Wed, Oct 7 · 4:45 p.m.

Math education

Math Tutor / Polya

LEADS

Eric Klavins · Dan Mikulincer

STUDENTS

Jake Malody

Hayden Guo

Amey Bharambe

Eshaan Kumar

AI Math Textbook

LEADS

Giovanni Inchiostro · Dhruv Bhatia

STUDENTS

Ethan Wang

Aarnith Netrawali

Saksham Singh

Nicole Ham

Abel Mesfin

Regular Meetings & Important Dates

Monday & Wednesday

4–5:30 p.m. · OUG 136

Projects are allowed to have different meeting times
but have to meet in-person at least once a week!

Social Event

Wednesday, November 4

Final presentations

Wednesday, December 9

Getting started

For project leaders

- Please gather your team members' GitHub usernames and polish your project README. (Fill the form to get your repo.)
- Think about what venue(s) you are going to submit your research!
- If you foresee any significant compute or other logistical requests, please send them to Michael via email. For AI subscriptions, fill out the form!

Research credit

SLN

23350

Faculty code

33548

Up to 2 credits

Discord & team requests

Lab Discord



discord.gg/Ge6BPRqrd

GitHub & AI team plans



[Open the team request form ↗](#)

Computing resources

AWS

Cloud servers (**EC2**) for experiments and hosting lab tools; object storage (**S3**) for datasets and results.

Request an instance or storage for your project.

[EC2 guide ↗](#)

Hyak

UW's shared computing cluster for CPU-intensive computation, parallel jobs, and moderate GPU workloads.

Use **Open OnDemand** in your browser, or SSH. **Slurm** schedules jobs.

[Getting started ↗](#)

Tillicum

UW's **NVIDIA H200 GPU** cluster for AI and GPU-intensive research.

Join a project allocation, activate your UW research-computing account, and run GPU jobs with **Slurm**.

[Getting started ↗](#)

Contact Michael for computing requests: zengrf@uw.edu.

Describe your project, CPU/GPU needs, memory, storage, and expected run time.

Publication venues

VENUE · OFFICIAL LINK	RESEARCH AREA	SUBMISSION DEADLINE
ICLR 2027	Representation learning	25 Sep 2026 · closed
ICML 2027	Machine learning	TBA · last: 28 Jan 2026 (closed)
NeurIPS 2027	ML, theory, benchmarks	TBA · last: 6 May 2026 (closed)
AISTATS 2027	Statistics & learning	6 Oct 2026 · abstracts already closed
COLM 2027	Language models	TBA · last: 31 Mar 2026 (closed)
COLT 2027	Learning theory	TBA
UAI 2027	Probabilistic reasoning	TBA · last: 25 Feb 2026 (closed)
AAAI 2027	AI, agents, reasoning	28 Jul 2026 · closed
IJCAI 2027	AI, agents, reasoning	11 Jan 2027 · abstract 4 Jan

Publication venues

VENUE · OFFICIAL LINK	RESEARCH AREA	SUBMISSION DEADLINE
TACAS 2027	Verification tools	15 Oct 2026
ESOP 2027	Programming languages	15 Oct 2026 · second round
CAV 2027	Computer-aided verification	20 Jan 2027
CADE-31 · 2027	Automated deduction	22 Feb 2027 · abstract 15 Feb
ITP 2027	Interactive theorem proving	TBA
CICM 2027	Computer mathematics	TBA
ISSAC 2027	Symbolic & algebraic computation	TBA
CPP 2027	Certified programs & proofs	10 Sep 2026 · closed

Publication venues

VENUE · OFFICIAL LINK	RESEARCH AREA	SUBMISSION DEADLINE
AI for Math · ICML 2026	AI for mathematics	25 May 2026 · closed
AI as a Tool for Math, CS & ML · ICML 2026	AI-assisted research	13 May 2026 · closed
MATH-AI · NeurIPS 2026	Mathematical reasoning	6 Sep 2026 · closed
Logical Reasoning of LLMs · ICLR 2026	Logic & reasoning	20 Feb / 20 Mar 2026 · closed
VerifAI-2 · ICLR 2026	AI & formal verification	8 Feb 2026 AoE · closed
Lean Workshop · FLoC 2026	Lean; talk proposals	22 Apr 2026 · closed
LFMTP 2026	Logical frameworks	1 May 2026 · closed
AITP 2027	AI & theorem proving	TBA

Publication venues

VENUE · OFFICIAL LINK	RESEARCH AREA	SUBMISSION DEADLINE
UniReps 2026 · Paris	Neural representations	10 Oct 2026 AoE · extended
Mechanistic Interpretability · ICML 2026	Circuits & mechanisms	8 May 2026 · closed
Interpretability as a Science · NeurIPS 2026	Methods & evaluation	1 Sep 2026 · closed
Interpretability for Discovery · NeurIPS 2026	Scientific discovery	2 Sep 2026 · closed
XAI4Science · NeurIPS 2026	Explainable AI in science	5 Sep 2026 · closed

Publication venues

VENUE · OFFICIAL LINK	RESEARCH AREA	SUBMISSION DEADLINE
NAACL / COLING 2027	NLP & language models	12 Oct 2026 ARR; commit 23 Dec
ACL 2027	NLP & language models	Jan 2027 ARR · exact day TBA
KDD 2027	Data science & AI	Feb 2027 cycle · exact day TBA
AIED 2027	AI in education	TBA · last: 2 Feb 2026 (closed)
EDM 2027	Educational data mining	TBA · last: 9 Feb 2026 (closed)
ICAIF 2027	AI in finance	TBA · last: 9 Aug 2026 (closed)
ICAIF 2026 workshops	Financial AI & agents	12 Oct 2026 AoE · extended

ARR = ACL Rolling Review. Submit for reviews, then commit the reviewed paper to a conference.

Publication venues

VENUE · OFFICIAL LINK	RESEARCH AREA	DEADLINE
TMLR · Transactions on ML Research	Machine learning	Rolling
JMLR · Journal of ML Research	Machine learning	Rolling
JAIR · Journal of AI Research	AI, agents & reasoning	Rolling
Journal of Automated Reasoning	Formal proofs & deduction	Rolling
Journal of Symbolic Computation	Computer algebra & algorithms	Rolling
Experimental Mathematics	Computation & mathematical discovery	Rolling
SIAM J. on Mathematics of Data Science	Mathematics of data science	Rolling
International J. of AI in Education	Learning & tutoring systems	Rolling
JOSS · Journal of Open Source Software	Research software & documentation	Rolling

Rolling = regular articles can be submitted throughout the year; special issues may have their own deadlines.

Discord & team requests

Lab Discord



discord.gg/Ge6BPRqrd

GitHub & AI team plans



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